Problem 1 **Fourier Series (10 points)** Consider the periodic function $f(t)$ with period 2π, defined by

$$f(t) = \begin{cases} t, & -\pi < t < 0, \\ \pi - t, & 0 < t < \pi, \\ 0, & t = 0, \pm\pi, \end{cases}$$

and extended periodically outside $-\pi < t < \pi$.

a) Determine whether $f(t)$ is even, odd, or neither. Explain your reasoning and discuss what this implies for the Fourier series of $f(t)$.

b) Compute the Fourier series of $f(t)$, giving explicit expressions for the Fourier coefficients a_0, a_n, b_n. Simplify as much as possible.

c) Write the first three nonzero terms of the Fourier series explicitly.

Solution:

a) Determine symmetry

Check $f(-t)$:

$$f(-t) = \begin{cases} -t, & 0 < t < \pi, \\ \pi + t, & -\pi < t < 0, \end{cases}$$

which is neither $f(t)$ nor $-f(t)$. Therefore, $f(t)$ is neither even nor odd, so the Fourier series will contain both cosine and sine terms.

b) Fourier series formula

For a 2π-periodic function:

$$f(t) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos(nt) + b_n \sin(nt)\right),$$

where

$$a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(t)\,dt, \quad a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(t)\cos(nt)\,dt, \quad b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(t)\sin(nt)\,dt.$$

Compute a_0:

$$a_0 = \frac{1}{\pi}\left(\int_{-\pi}^{0} t\,dt + \int_{0}^{\pi}(\pi - t)\,dt\right) = \frac{1}{\pi}\left(0 - \frac{\pi^2}{2} + \frac{\pi^2}{2}\right) = 0$$

Compute a_n:

$$a_n = \frac{1}{\pi}\left(\int_{-\pi}^{0} t\cos(nt)\,dt + \int_{0}^{\pi}(\pi - t)\cos(nt)\,dt\right)$$

$$\int t\cos(nt)\,dt = \frac{t\sin(nt)}{n} + \frac{\cos(nt)}{n^2}$$

$$\int_{-\pi}^{0} t\cos(nt)\,dt = \frac{1-(-1)^n}{n^2}, \quad \int_{0}^{\pi}(\pi - t)\cos(nt)\,dt = \frac{1-(-1)^n}{n^2}$$

$$\Rightarrow a_n = \frac{2(1-(-1)^n)}{\pi n^2}$$

$$\text{Thus: } a_n = \begin{cases} 0, & n \text{ even}, \\ \dfrac{4}{\pi n^2}, & n \text{ odd}. \end{cases}$$

Compute b_n:

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(t)\sin(nt)\,dt = \frac{1}{\pi}\left(\int_{-\pi}^{0} t\sin(nt)\,dt + \int_{0}^{\pi}(\pi - t)\sin(nt)\,dt\right).$$

First integral:

$$\int t\sin(nt)\,dt = -\frac{t\cos(nt)}{n} + \frac{\sin(nt)}{n^2}.$$

Hence

$$\int_{-\pi}^{0} t\sin(nt)\,dt = \left[-\frac{t\cos(nt)}{n} + \frac{\sin(nt)}{n^2}\right]_{t=-\pi}^{0} = \frac{\pi\cos(n\pi)}{n}.$$

Second integral:

$$\int_0^\pi (\pi - t)\sin(nt)\,dt = \pi\int_0^\pi \sin(nt)\,dt - \int_0^\pi t\sin(nt)\,dt.$$

We have

$$\int_0^\pi \sin(nt)\,dt = \frac{1-(-1)^n}{n},$$

and

$$\int_0^\pi t\sin(nt)\,dt = \left[-\frac{t\cos(nt)}{n} + \frac{\sin(nt)}{n^2}\right]_0^\pi = -\frac{\pi\cos(n\pi)}{n}.$$

Thus

$$\int_0^\pi (\pi - t)\sin(nt)\,dt = \frac{\pi(1-(-1)^n)}{n} + \frac{\pi\cos(n\pi)}{n}.$$

Adding and dividing by π,

$$b_n = \frac{1}{\pi}\left(\frac{\pi\cos(n\pi)}{n} + \frac{\pi(1-(-1)^n)}{n} + \frac{\pi\cos(n\pi)}{n}\right) = \frac{1-(-1)^n}{n}.$$

Therefore

$$\boxed{b_n = \frac{1-(-1)^n}{n} = \begin{cases} 0, & n \text{ even}, \\ \dfrac{2}{n}, & n \text{ odd}. \end{cases}}$$

The Fourier series becomes

$$f(t) \sim \sum_{\substack{n\geq 1 \\ n \text{ odd}}} \frac{4}{\pi n^2}\cos(nt) \; + \; \sum_{\substack{n\geq 1 \\ n \text{ odd}}} \frac{2}{n}\sin(nt).$$

c) First three nonzero terms

$$\boxed{f(t) \approx \frac{4}{\pi}\cos t + 2\sin t + \frac{4}{9\pi}\cos 3t + \frac{2}{3}\sin 3t + \cdots}.$$

Grading manual: a) 1 point, b) 8 points c) 1 point

Problem 2 Interpolation (10 points)
The following table shows the values of a function $f(x)$ at the following points:

x	$f(x)$
0	0
1	2
2	6
3	12

a) Compute an interpolating polynomial $p(x)$ of minimal necessary degree for the given data. Simplify the resulting polynomial as much as possible.

b) Use $p(x)$ to compute an approximate value of $f(1.5)$.

c) Briefly explain why the interpolating polynomial of minimal necessary degree exists and is unique for this data set.

Solution:

a) **Using Lagrange:** We are given data points (x_i, f_i) for $i = 0, 1, 2, 3$:

$$(0,0), \quad (1,2), \quad (2,6), \quad (3,12).$$

The Lagrange interpolant of degree at most 3 is

$$p(x) = \sum_{i=0}^{3} f_i\, \ell_i(x), \qquad \ell_i(x) = \prod_{\substack{j=0 \\ j\neq i}}^{3} \frac{x - x_j}{x_i - x_j}.$$

With nodes $x_0 = 0,\ x_1 = 1,\ x_2 = 2,\ x_3 = 3$, the basis polynomials are

$$\begin{aligned}
\ell_0(x) &= \frac{(x-1)(x-2)(x-3)}{(0-1)(0-2)(0-3)} = -\frac{1}{6}(x-1)(x-2)(x-3),\\
\ell_1(x) &= \frac{x(x-2)(x-3)}{(1-0)(1-2)(1-3)} = \frac{1}{2}x(x-2)(x-3),\\
\ell_2(x) &= \frac{x(x-1)(x-3)}{(2-0)(2-1)(2-3)} = -\frac{1}{2}x(x-1)(x-3),\\
\ell_3(x) &= \frac{x(x-1)(x-2)}{(3-0)(3-1)(3-2)} = \frac{1}{6}x(x-1)(x-2).
\end{aligned}$$

Thus

$$p(x) = 0 \cdot \ell_0(x) + 2\,\ell_1(x) + 6\,\ell_2(x) + 12\,\ell_3(x).$$

Expanding and simplifying gives

$$\boxed{p(x) = x^2 + x.}$$

Using Newton:

We use Newton's divided difference method.

Since four interpolation points are given, we are a priori looking for an interpolating polynomial of degree at most 3.

Zeroth-order divided differences:

$$f[x_0] = 0, \quad f[x_1] = 2, \quad f[x_2] = 6, \quad f[x_3] = 12.$$

First-order divided differences:

$$\begin{aligned} f[x_0, x_1] &= \frac{2-0}{1-0} = 2, \\ f[x_1, x_2] &= \frac{6-2}{2-1} = 4, \\ f[x_2, x_3] &= \frac{12-6}{3-2} = 6. \end{aligned}$$

Second-order divided differences:

$$\begin{aligned} f[x_0, x_1, x_2] &= \frac{4-2}{2-0} = 1, \\ f[x_1, x_2, x_3] &= \frac{6-4}{3-1} = 1. \end{aligned}$$

Third-order divided difference:

$$f[x_0, x_1, x_2, x_3] = \frac{1-1}{3-0} = 0.$$

Since the third-order divided difference is zero, the interpolating polynomial has degree at most 2.

The Newton form of the interpolating polynomial is therefore

$$\begin{aligned} p_3(x) = f[x_0] + f[x_0, x_1](x - x_0) + f[x_0, x_1, x_2](x - x_0)(x - x_1) \\ + f[x_0, x_1, x_2, x_3](x - x_0)(x - x_1)(x - x_2). \end{aligned}$$

Substitute the computed values:

$$p(x) = 0 + 2(x - 0) + 1(x - 0)(x - 1).$$

Simplify:

$$p(x) = 2x + x(x - 1) = 2x + x^2 - x = x^2 + x.$$

Thus, the interpolating polynomial is:

$$\boxed{p(x) = x^2 + x.}$$

Using the direct approach:

Let us consider a quadratic polynomial

$$p(x) = ax^2 + bx + c,$$

and determine a, b, c directly from the pointwise conditions.

From $p(0) = 0$ we get

$$c = 0.$$

From $p(1) = 2$ we get

$$a + b + c = 2 \quad \Longrightarrow \quad a + b = 2.$$

From $p(2) = 6$ we get

$$4a + 2b + c = 6 \quad \Longrightarrow \quad 4a + 2b = 6.$$

Using $c = 0$, solve the linear system

$$\begin{aligned} a + b &= 2, \\ 4a + 2b &= 6. \end{aligned}$$

Multiply the first equation by 2 and subtract from the second:

$$4a + 2b - 2(a + b) = 6 - 4 \quad \Longrightarrow \quad 2a = 2 \quad \Longrightarrow \quad a = 1.$$

Then $b = 2 - a = 1$. Thus

$$\boxed{p(x) = x^2 + x.}$$

b) **Estimate at** $x = 1.5$**:**

$$f(1.5) \approx p(1.5) = (1.5)^2 + 1.5 = 2.25 + 1.5 = 3.75.$$

c) **Existence and Uniqueness:** For **distinct** nodes $x_0, \ldots, x_3$, there exists a unique polynomial of degree ≤ 3 that interpolates the data.

If q were another such polynomial, then $r = p - q$ would be a nonzero polynomial of degree ≤ 3 with four distinct roots $x_0, \ldots, x_3$, which is impossible. Hence p is unique.

Grading manual: a) 7 points b) 1 point c) 2 points
Remark for c): Award **1 point** if the student provides any of the following valid arguments:

- References to the Lagrange interpolation formula as a constructive proof of existence. (e.g. "The Lagrange form already gives a polynomial that interpolates the data.")
- References to the standard theorem from the lecture:
(e.g. "For $n + 1$ distinct nodes, there exists a polynomial of degree at most n that interpolates the data.")
- Mentions that the nodes $x_0, \ldots, x_3$ are distinct.

No point if:

- "Existence" is merely stated without any explanation.
- The argument is incorrect or irrelevant.

Award **1 point** for a correct explanation of the standard uniqueness argument. For example:

- "If p and q both interpolate the data, then $r = p - q$ is a polynomial of degree at most 3 with four distinct zeros. This is impossible unless $r \equiv 0$, so $p = q$."

No point if:

- "Uniqueness" is only stated without justification.
- The reasoning is logically incorrect.

Problem 3 **Laplace Transform (10 points)**

a) Compute the Laplace transform of the function

$$f(t) = e^{-t}(1 + 2t).$$

b) Using your result from a) and the **convolution theorem**, compute the inverse Laplace transform of

$$Y(s) = \frac{1}{s+1} \cdot \mathcal{L}\{f(t)\},$$

where $f(t)$ is given in a).

c) Solve the initial value problem

$$y'(t) + y(t) = e^{-t}(1 + 2t), \quad y(0) = 0.$$

(Hint: use your result from b).)

Solution

a) **Using the definition of the Laplace transform:** Compute the Laplace transform of $f(t) = e^{-t}(1 + 2t)$:

$$\mathcal{L}\{f(t)\} = \int_0^\infty e^{-st}e^{-t}(1 + 2t)\, dt = \int_0^\infty e^{-(s+1)t}(1 + 2t)\, dt.$$

Split into two integrals:

$$\int_0^\infty e^{-(s+1)t} dt = \left[-\frac{1}{s+1}e^{-(s+1)t}\right]_0^\infty = 0 - \left(-\frac{1}{s+1}\right) = \frac{1}{s+1}.$$

For the second integral, we have

$$\int_0^\infty 2te^{-(s+1)t} dt = 2\int_0^\infty te^{-(s+1)t} dt.$$

Use integration by parts:

$$\int_0^\infty te^{-(s+1)t} dt = \left[-\frac{t}{s+1}e^{-(s+1)t}\right]_0^\infty + \frac{1}{s+1}\int_0^\infty e^{-(s+1)t} dt$$

Evaluate the boundary term:

$$\lim_{t\to\infty} \frac{t}{s+1}e^{-(s+1)t} = 0, \quad \text{and at } t = 0 : -\frac{0}{s+1}e^{0} = 0$$

The remaining integral:

$$\frac{1}{s+1}\int_0^\infty e^{-(s+1)t}dt = \frac{1}{s+1}\cdot\frac{1}{s+1} = \frac{1}{(s+1)^2}$$

Multiply by 2:

$$2\int_0^\infty te^{-(s+1)t}dt = \frac{2}{(s+1)^2}$$

Combine both integrals:

$$\boxed{\mathcal{L}\{f(t)\} = \frac{1}{s+1} + \frac{2}{(s+1)^2}}$$

Using the Laplace transform of $1 + 2t$ und using the shifting theorem:
We know the basic Laplace transforms:

$$\mathcal{L}\{1\} = \frac{1}{s}, \quad \mathcal{L}\{t\} = \frac{1}{s^2}.$$

By linearity:

$$\mathcal{L}\{1+2t\} = \mathcal{L}\{1\} + 2\mathcal{L}\{t\} = \frac{1}{s} + \frac{2}{s^2}.$$

Let

$$G(s) = \mathcal{L}\{1+2t\} = \frac{1}{s} + \frac{2}{s^2}.$$

The first shift theorem states:

$$\mathcal{L}\{e^{at}g(t)\} = G(s-a).$$

Here, $a = -1$, so

$$\mathcal{L}\{e^{-t}(1+2t)\} = G(s+1),$$

and therefore

$$G(s+1) = \frac{1}{s+1} + \frac{2}{(s+1)^2}.$$

Hence,

$$\boxed{\mathcal{L}\{e^{-t}(1+2t)\} = \frac{1}{s+1} + \frac{2}{(s+1)^2}}.$$

b) Compute the inverse Laplace transform using the convolution theorem:

$$Y(s) = \frac{1}{s+1} \cdot \mathcal{L}\{f(t)\} \overset{a)}{=} \frac{1}{s+1} \cdot \left(\frac{1}{s+1} + \frac{2}{(s+1)^2} \right)$$

$$\overset{\text{convolution theorem}}{\Longrightarrow} \mathcal{L}^{-1}\{Y(s)\} = \mathcal{L}^{-1}\left\{\frac{1}{s+1}\right\} * f(t) = \int_0^t e^{-(t-\tau)} f(\tau)\, d\tau.$$

Substitute $f(\tau) = e^{-\tau}(1+2\tau)$:

$$y_1(t) = \int_0^t e^{-(t-\tau)} \cdot e^{-\tau}(1+2\tau)\, d\tau = \int_0^t e^{-t}(1+2\tau)\, d\tau = e^{-t} \int_0^t (1+2\tau)\, d\tau.$$

Compute the integral:

$$\int_0^t (1+2\tau)\, d\tau = t + t^2.$$

Thus,

$$\mathcal{L}^{-1}\{Y(s)\} = e^{-t}(t+t^2).$$

c) Solve the IVP:

Take Laplace transforms:

$$\mathcal{L}\{y'\} + \mathcal{L}\{y\} = \mathcal{L}\{f(t)\} \implies (sY(s) - y(0)) + Y(s) = \mathcal{L}\{f(t)\}.$$

Using $y(0) = 0$:

$$(s+1)Y(s) = \mathcal{L}\{f(t)\} \implies Y(s) = \frac{1}{s+1} \cdot \mathcal{L}\{f(t)\}.$$

From part b):

$$y(t) = \mathcal{L}^{-1}\{Y(s)\} = e^{-t}(t+t^2).$$

Grading manual

a) (4p)

b) (4p)

c) (2p)

Problem 4 Numerical integration (10 points)

a) Consider the quadrature rule

$$\int_{-1}^{1} f(s)ds \approx Q[f] = w_0 f(-1) + w_1 f\left(-\frac{1}{3}\right) + w_2 f\left(\frac{1}{3}\right) + w_3 f(1).$$

Assume that it is exact for all polynomials of degree 3 or lower.

Deduce the values of w_0, w_1, w_2, w_3 and show that

$$w_0 + w_1 = 1 \quad \text{and} \quad 9w_0 + w_2 = 3.$$

b) Use Taylor expansion of the function f at the point $x_k + \frac{1}{2}h$ to derive the midpoint rule of integration

$$\int_{x_k}^{x_k+h} f(x)\,dx = hf(x_k + \tfrac{1}{2}h) + \tfrac{1}{24}h^3 f''(\tilde{\xi}), \qquad x_k < \tilde{\xi} < x_k + h.$$

Hint: You can use the mean value theorem for integrals: If $g(x)$ is continuous and $w(x) \geq 0$ for all $x \in [a,b]$, then

$$\exists\,\xi \in [a,b] \text{ such that } \int_a^b g(x)\,w(x)\,dx = g(\xi)\int_a^b w(x)\,dx.$$

Solution

a) The formula is exact for all polynomials of degree 3 or lower if, and only if, it is exact for x^0, x^1, x^2, x^3:

$$\begin{aligned} 2 &= w_0 + w_1 + w_2 + w_3 \\ 0 &= -w_0 - \tfrac{1}{3}w_1 + \tfrac{1}{3}w_2 + w_3 \\ \tfrac{2}{3} &= w_0 + \tfrac{1}{9}w_1 + \tfrac{1}{9}w_2 + w_3 \\ 0 &= -w_0 - \tfrac{1}{27}w_1 + \tfrac{1}{27}w_2 + w_3. \end{aligned}$$

From the 2nd equation and the 4th equation we see that $w_0 = w_3$ and $w_1 = w_2$. The 1st equation and the 3rd equation then become

$$\begin{aligned} 2 &= 2w_0 + 2w_1, \\ \tfrac{2}{3} &= 2w_0 + \tfrac{2}{9}w_1. \end{aligned}$$

This leads to

$$w_0 = w_3 = \tfrac{1}{4}, \qquad w_1 = w_2 = \tfrac{3}{4}.$$

b) Recall Taylor's expansion at the point a:

$$f(x) = \sum_{i=0}^{n-1} \frac{f^{(i)}(a)}{i!}(x-a)^i + \frac{f^{(n)}(\xi(x))}{n!}(x-a^n)$$

where $\xi(x) \in]u,\, v[$.

Then the Taylor expansion of $f(x)$ at the point $a = x_k + \frac{1}{2}h$ (up to 2nd order) gives

$$f(x) = f(x_k + \tfrac{1}{2}h) + f'(x_k + \tfrac{1}{2}h)(x - (x_k + \tfrac{1}{2}h)) + \frac{1}{2} f''(\xi(x))(x - (x_k + \tfrac{1}{2}h))^2,$$

where $\xi(x) \in [x_k,\, x_k + h]$.

Now, integrating on both sides gives

$$\begin{aligned}\int_{x_k}^{x_k+h} f(x)dx = hf(x_k + \tfrac{1}{2}h) + f'(x_k + \tfrac{1}{2}h) \int_{x_k}^{x_k+h} (x - (x_k + \tfrac{1}{2}h))dx \\ + \tfrac{1}{2} \int_{x_k}^{x_k+h} (x - (x_k + \tfrac{1}{2}h))^2 f''(\xi(x))dx.\end{aligned}$$

An explicit computation shows that the first integral on the right vanishes:

$$\int_{x_k}^{x_k+h} (x - (x_k + \tfrac{1}{2}h))dx = 0.$$

To the last integral on the right we apply the mean value theorem for integrals:

$$\int_{x_k}^{x_k+h} (x - (x_k + \tfrac{1}{2}h))^2 f''(\xi(x))dx = f''(\tilde{\xi}) \int_{x_k}^{x_k+h} (x - (x_k + \tfrac{1}{2}h))^2 dx,$$

for $x_k < \tilde{\xi} < x_k + h$, giving

$$\int_{x_k}^{x_k+h} f(x)dx = hf(x_k + \tfrac{1}{2}h) + \tfrac{1}{2} f''(\tilde{\xi}) \int_{x_k}^{x_k+h} (x - (x_k + \tfrac{1}{2}h))^2 dx,$$

for $x_k < \tilde{\xi} < x_k + h$.

Another explicit computation gives the value of the integral on the right

$$\int_{x_k}^{x_k+h} (x - (x_k + \tfrac{1}{2}h))^2 dx = \tfrac{h^3}{12}.$$

Thus, we find

$$\int_{x_k}^{x_k+h} f(x)\, dx = hf(x_k + \tfrac{1}{2}h) + \tfrac{h^3}{24} f''(\tilde{\xi}) = (x_{k+1} - x_k) f\left(\frac{x_k + x_{k+1}}{2}\right) + \tfrac{h^3}{24} f''(\tilde{\xi}),$$

for $x_k < \tilde{\xi} < x_k + h$.

Grading manual

a) (4p)

b) (6p)

Problem 5 Wave equation and Heat equation (10 points)

a) Consider the function

$$u(x,t) = \sin(2t)\cos(x)$$

and verify that it is a solution to the wave equation

$$u_{tt} = 4u_{xx}.$$

b) Let $c_1, \ldots, c_n$ denote real constants and consider the modified wave equation

$$u_{tt} = \sum_{i=1}^{n} c_i^2 u_{x_i x_i}. \tag{1}$$

Let F be an arbitrary twice continuously differentiable function, meaning its second derivative exists and is continuous. Show that for an arbitrary vector $a = (a_1, \ldots, a_n)$ the function $u(x,t) = u(x_1, \ldots, x_n, t)$

$$u(x,t) = F(a \cdot x - \lambda t)$$

is a solution of (1) if

$$\lambda^2 - \sum_{i=1}^{n} c_i^2 a_i^2 = 0$$

c) Find the solution to the boundary value problem

$$u_t(x,t) - u_{xx}(x,t) = 0, \quad 0 < x < \pi, \quad t > 0$$

$$i)\ u(0,t) = 0 = u(\pi,t), \quad t > 0$$
$$ii)\ u(x,0) = 100, \quad 0 < x < \pi.$$

Note: Here you can use the formula for the solution of the heat equation derived in the lecture:

$$u(x,t) = \sum_{n=1}^{\infty} b_n e^{-\lambda_n^2 t} \sin\left(\frac{n\pi}{L}x\right), \tag{2}$$

where

$$b_n = \frac{2}{L}\int_0^L f(x)\sin\left(\frac{n\pi}{L}x\right)dx, \qquad \lambda_n = c\frac{n\pi}{L}, \qquad n = 1, 2, \ldots \tag{3}$$

You do <u>not</u> need to derive the formula yourself, but you need to apply it.

Solution

a) We compute partial derivations:

$$u_t(x,t) = \partial_t u(x,t) = 2\cos(2t)\cos(x)$$

and

$$u_{tt}(x,t) = \partial_t\partial_t u(x,t) = -4\sin(2t)\cos(x).$$

Now with respect to x:

$$u_x(x,t) = \partial_x u(x,t) = -\sin(2t)\sin(x)$$

and

$$u_{xx}(x,t) = \partial_x\partial_x u(x,t) = -\sin(2t)\cos(x).$$

We see that:

$$u_{tt} = 4u_{xx}.$$

b) We compute

$$u_{tt}(x,t) = \partial_t\partial_t F(a\cdot x - \lambda t) = \lambda^2 F''(a\cdot x - \lambda t)$$

and

$$u_{x_i x_i}(x,t) = \partial_{x_i}\partial_{x_i} F(a\cdot x - \lambda t) = a_i^2 F''(a\cdot x - \lambda t).$$

We then see that the condition $\lambda^2 - \sum_{i=1}^n c_i^2 a_i^2 = 0$ implies that $u(x,t) = F(a\cdot x - \lambda t)$ is a solution to (1).

c) Recall that the solution to the 1–dimensional heat boundary value problem

$$u_t(x,t) - c^2 u_{xx}(x,t) = 0, \quad 0 < x < L, \quad t > 0$$

$$i)\ u(0,t) = 0 = u(L,t), \quad t > 0$$
$$ii)\ u(x,0) = f(x), \quad 0 < x < L$$

is given by

$$u(x,t) = \sum_{n=1}^{\infty} b_n e^{-\lambda_n^2 t} \sin\left(\frac{n\pi}{L}x\right), \tag{4}$$

where

$$b_n = \frac{2}{L}\int_0^L f(x)\sin\left(\frac{n\pi}{L}x\right)dx, \qquad \lambda_n = c\frac{n\pi}{L}, \qquad n = 1,2,\dots \tag{5}$$

From (4) and $L = \pi$ and $c = 1$, we deduce that

$$u(x,t) = \sum_{n=1}^{\infty} b_n e^{-n^2 t}\sin nx \tag{6}$$

where

$$b_n = \frac{2}{\pi}\int_0^{\pi} 100\sin nx\,dx = \frac{200}{n\pi}(1-\cos n\pi).$$

Substituting b_n and using the fact that $(1-\cos n\pi) = 0$ if n is even and 2 if n is odd, we obtain

$$\begin{aligned} u(x,t) &= \frac{400}{\pi}\sum_{\substack{n=1 \\ n \text{ odd}}}^{\infty} \frac{e^{-n^2 t}}{n}\sin nx \\ &= \frac{400}{\pi}\sum_{k=0}^{\infty} \frac{e^{-(2k+1)^2 t}}{2k+1}\sin(2k+1)x. \end{aligned}$$

Interpretation: If we fix a value of time t in the series solution, then we obtain a function of x. It describes the temperature distribution of the bar at the given time t. In particular, when $t = 0$, $u(x,0)$ gives the sine-expansion of the initial temperature distribution $f(x)$ over half-range domain.

Grading manual

a) (2p)

b) (3p)

c) (5p)

Problem 6 Numerical methods for ODEs (10 points)

a) Rewrite the following initial value problem (IVP) as a first order system with initial value.

$$x'''(t) - 2x''(t) - x'(t) + 2x(t) = 1 - 2t, \quad x(0) = x'(0) = 1,\ x''(0) = 0.$$

b) Consider the first order IVP

$$x'(t) = 1 + t - x(t), \quad t > 0, \quad x(0) = x_0 = 0.$$

i) Use the explicit Euler method with a general step size h to compute the first three approximations $x_1, x_2, x_3, \ldots$.

ii) Deduce an expression for x_n in terms of $t_n = nh$.

iii) Use the result to guess the exact solution of the IVP.

c) Show that the Runge–Kutta method with Butcher tableau

$$\begin{array}{c|cc} 0 & 0 & 0 \\ 1 & \frac{1}{2} & \frac{1}{2} \\ \hline & \frac{1}{2} & \frac{1}{2} \end{array}$$

is equivalent to the trapezoidal rule

$$x_{n+1} = x_n + \tfrac{1}{2}h\Big(f(t_{n+1}, x_{n+1}) + f(t_n, x_n)\Big).$$

Solution

a) We denote the n-th derivative of the function $x(t)$ by $x^{(n)}(t) := \frac{d^n}{dt^n}x(t)$. Now, we define the functions

$$u_i(t) := x^{(i-1)}(t), \quad i = 1, 2, 3.$$

We see that

$$u_1'(t) = u_2(t), \quad u_2'(t) = u_3(t),$$

such that

$$\frac{d}{dt}\begin{pmatrix} u_1(t) \\ u_2(t) \\ u_3(t) \end{pmatrix} = \begin{pmatrix} u_2(t) \\ u_3(t) \\ 2u_3(t) + u_2(t) - 2u_1(t) + 1 - 2t \end{pmatrix}$$
$$= \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -2 & 1 & 2 \end{pmatrix}\begin{pmatrix} u_1(t) \\ u_2(t) \\ u_3(t) \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ 1 - 2t \end{pmatrix}.$$

and initial value
$$\begin{pmatrix} u_1(0) \\ u_2(0) \\ u_3(0) \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}.$$

b) We consider the IVP $x'(t) = 1 + t - x(t), \quad t > 0, \quad x(0) = x_0 = 0$ which we write as $x'(t) = f(t, x), \quad t > 0, \quad x(0) = x_0 = 0$, with $f(t, x) := 1 + t - x(t)$. Euler's method: $x_0 = 0$

$$x_{n+1} = x_n + hf(t_n, x_n) = x_n + h(1 + t_n - x_n), \quad n \geq 0, \quad t_n = nh. \tag{7}$$

i) Then
$$x_1 = h, \quad x_2 = 2h, \quad x_3 = 3h, \ldots$$

ii) This suggests to consider $x_n = nh = t_n$. It can be verified by induction using (7).

iii) It is natural to assume that the exact solution of the IVP is $x(t) = t$. This can be checked by substituting it into $x'(t) = 1 + t - x(t), \quad t > 0, \quad x(0) = x_0 = 0$.

c) From the Butcher tableau we deduce the steps and final update equation

$$\begin{aligned} K_1 &= f(t_n, x_n) \\ K_2 &= f(t_n + h, x_n + \tfrac{1}{2}h(K_1 + K_2)) \\ x_{n+1} &= x_n + \tfrac{1}{2}h(K_1 + K_2). \end{aligned}$$

From the last two equations we see that
$$K_2 = f(t_{n+1}, x_{n+1}).$$

This then implies that
$$x_{n+1} = x_n + \tfrac{1}{2}h(f(t_n, x_n) + f(t_{n+1}, x_{n+1}))$$

which is just the Trapezoidal rule.

Grading manual

a) (2p)

b) (3p)

c) (5p)

Grading Scale (The following grading scheme is adjusted from this one: https://i.ntnu.no/wiki/-/wiki/English/Grading+scale+using+percentage+points):

A	54–60
B	47–53
C	40–46
D	33–39
E	25–32
F	0–24